

Matter Library Integration Plan v2.0

Physically Based MaterialX Pipeline for Blender 5.1 + Unreal Engine (Substrate)

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Status: Implementation Phase Ready

Vision and Goals

The Matter Library Project aims to create a unified physically-based material system that produces visually identical results across real-time and offline render engines. Instead of optimizing for low-poly games, this library is built for lifelike, physically scaled, real-world materials usable in cinematic, virtual-production, and real-time visualization contexts.

Key Principles

1. MaterialX as single source of truth — authored once in open form.
2. Full physical scale — every texture's meters-per-tile defined.
3. No visual fakery — geometry holds real form.
4. Cross-engine visual parity — Blender 5.1 and Unreal 5.6+.
5. Open, extensible system — community contributions encouraged.

Architecture Overview

MaterialX serves as the canonical format, bridged to Blender's Principled BSDF and Unreal Engine's Substrate. This enables identical physical response across both real-time and offline contexts. Five master materials govern the look: Opaque, Masked, TranslucentThin, TwoLayer, and Emissive.

Library Foundations

145 Matter entries organized into 38 Families, grouped by MetaOrigin (Natural, Synthetic, Engineered, Environmental, Utility). Each has defined real-world texture scale and assigned master material. Texture sets include overlays and masks, each RGBA packed for reusability.

MaterialX Canonical Structure

Each .mtlx defines metadata (family, origin, scale, master) and nodegraphs for base PBR, overlays, and masks. Overlays use RGBA channels (Normal XY / Roughness Bias / Mask / AO). MaskSets

encode up to four semantic channels for layered materials.

Interchange Bridges

Blender: MaterialX → Principled BSDF with Master Node Groups and correct texture color spaces.

Unreal: MaterialX → Substrate Material Instances referencing prebuilt master templates. Metadata stored as asset tags. Lumen lighting ensures accurate PBR output.

Runtime Strategy

Unreal runtime manipulates material parameters only—no reimporting .mtlx. Deterministic filenames support asset lookup. Build phase converts .mtlx to UE instances. Supports high-fidelity 8K texture sources downsampled for performance.

Visual Parity Workflow

Author in MaterialX Viewer → Validate in Blender → Verify in Unreal → Automate comparisons ($\Delta E < 2$). Each material behaves consistently under identical HDR environments and exposure settings.

Naming & File Standards

Naming pattern: __-__clean_v#### (e.g., engineered_metal_steel-brushed_clean_v001). Structured folders for natural/synthetic/engineered groups with shared overlays and masksets.

Open Source Alignment

Based on ASWF MaterialX BSD license. Tooling via MaterialX Python API and Unreal conversion scripts. Goal: public release of a canonical Matter Library usable across DCCs and render engines.

Roadmap

Phase 1 – Core Setup ■

Phase 2 – MaterialX Schema v2

Phase 3 – Engine Templates

Phase 4 – Library Buildout

Phase 5 – Conversion Pipeline

Phase 6 – Visual Parity QA

Phase 7 – Open Release

Design Rationale Summary

MaterialX chosen for openness and physics correctness. Five master materials ensure simplicity. LCD parameters enforce cross-engine consistency. Overlays/masks reduce redundancy. 8K sources future-proof fidelity. Deterministic naming and metadata guarantee automation.

Future Extensions

- Character materials with dual-lobe SSS/specular.
- Procedural variants (wet/dusty/worn).
- AI-based classification and tagging.
- Expansion to Omniverse, Houdini, Arnold, and USD pipelines.

■ Summary

We now have a coherent taxonomy, unified master material architecture, defined MaterialX extensions, and a cross-engine runtime plan. This blueprint enables physically accurate, reproducible materials—rendered consistently from Blender to Unreal—with open-source alignment and future extensibility.